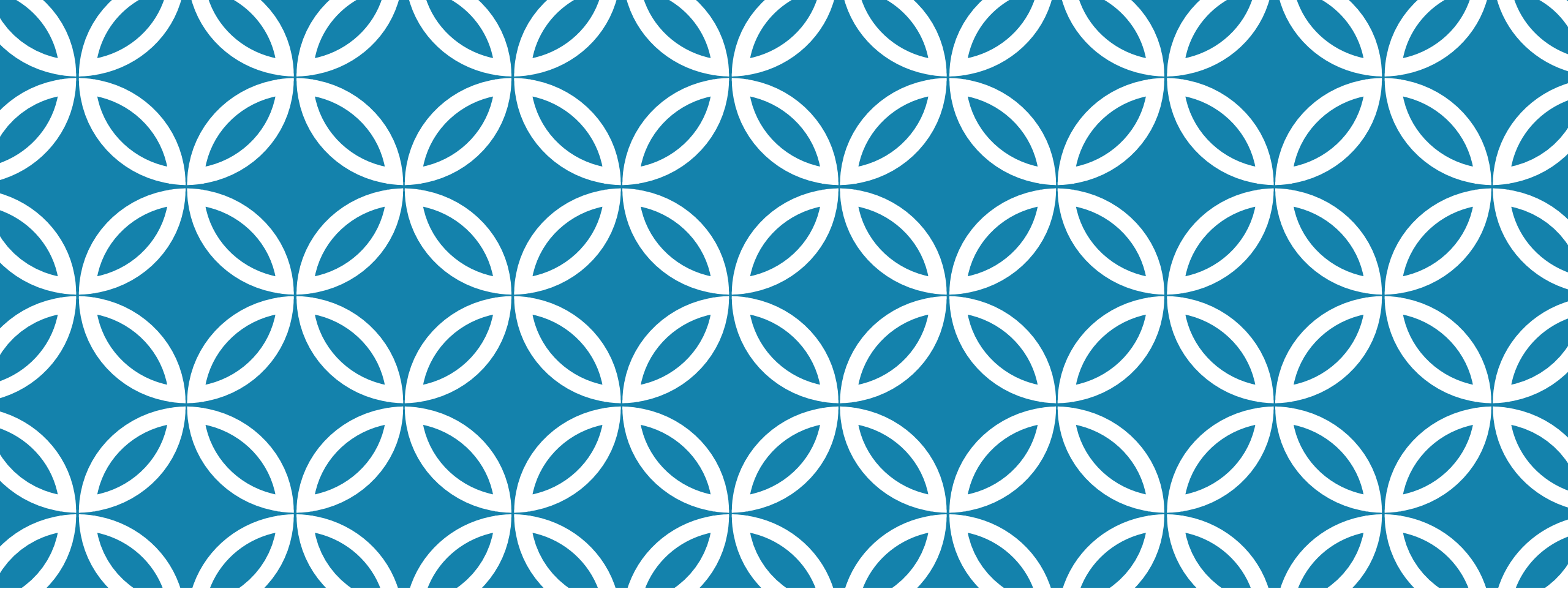


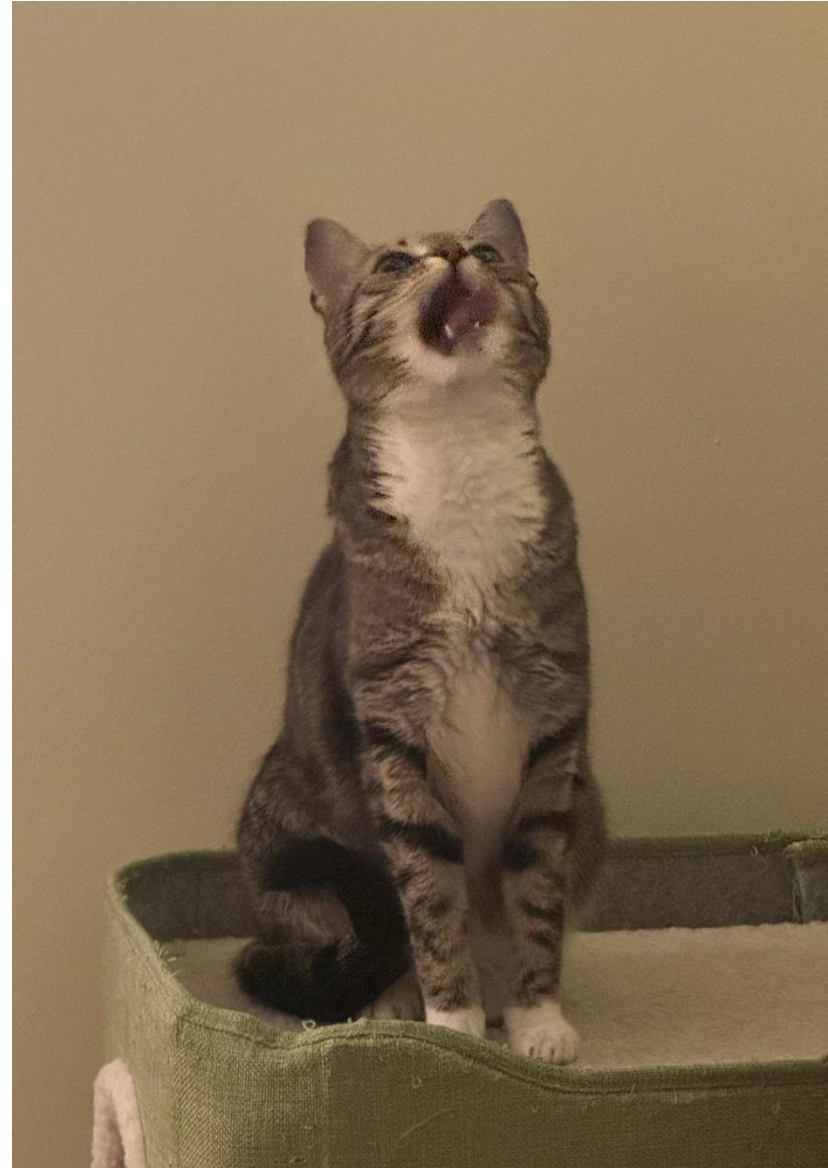


OPTICAL CHARACTERIZATION OF BGO CRYSTALS FOR PRECISION CALORIMETRY IN FUTURE COLLIDER EXPERIMENTS

Bhavya Singhal
PHYS 675 – Final Project

OUTLINE

- ❖ Introduction and Motivation
- ❖ Characterization Strategy
- ❖ Crystal Characterization Tests
- ❖ Results
- ❖ Current work and Future Plans
- ❖ Summary



WHY CALORIMETRY MATTERS?

Calorimeters are the backbone of modern HEP detectors:

- Measure particle energy with high precision
- Provide jet reconstruction, missing energy, particle ID
- FCC-ee/hh requires **unprecedented energy resolution**
- Calorimeters are the ideal instrument to measure the full energy of particles, particularly at high momentum

Core challenge:

- Most calorimeters suffer from **large event-to-event fluctuations** in the electromagnetic fraction → limits resolution.

LIMITATIONS OF EXISTING CALORIMETER TECHNOLOGIES

- Hadronic showers involve complex inelastic interactions with nuclei
- Energy Lost to:
 - Nuclear breakup (binding energy)
 - Neutrons escaping or depositing energy late
 - Heavy fragments, neutrinos, slow secondaries
- This “invisible energy” causes event-by-event fluctuations.
- Leads to poor hadronic resolution in conventional calorimeters:

$$\sigma_E / E \approx 50\% / \sqrt{E}$$

MAXIMAL-INFORMATION CALORIMETRY

- Invisible energy correlates with:
 - Number of nuclear breakups
 - Amount of non-relativistic (slow) charged particles
- Measuring *additional observables* allows shower-by-shower correction.
- Key idea: **If we can estimate the EM fraction f_{em} , we can correct for the missing energy.**
- Dual-readout calorimetry is one powerful implementation of this idea.

DUAL-READOUT CALORIMETRY (THE SOLUTION)

Key idea: measure *two independent signals*

- **Cherenkov light:** proportional to EM component
- **Scintillation light:** proportional to total deposited energy

Benefits:

- Event-by-event correction of EM fraction
- Drastically reduces non-compensation
- Achievable hadronic energy resolution: $\sim 15\%/\sqrt{E} \rightarrow 5\%/\sqrt{E}$

Therefore:

- We need crystals that support **clear separation** of the two light components.

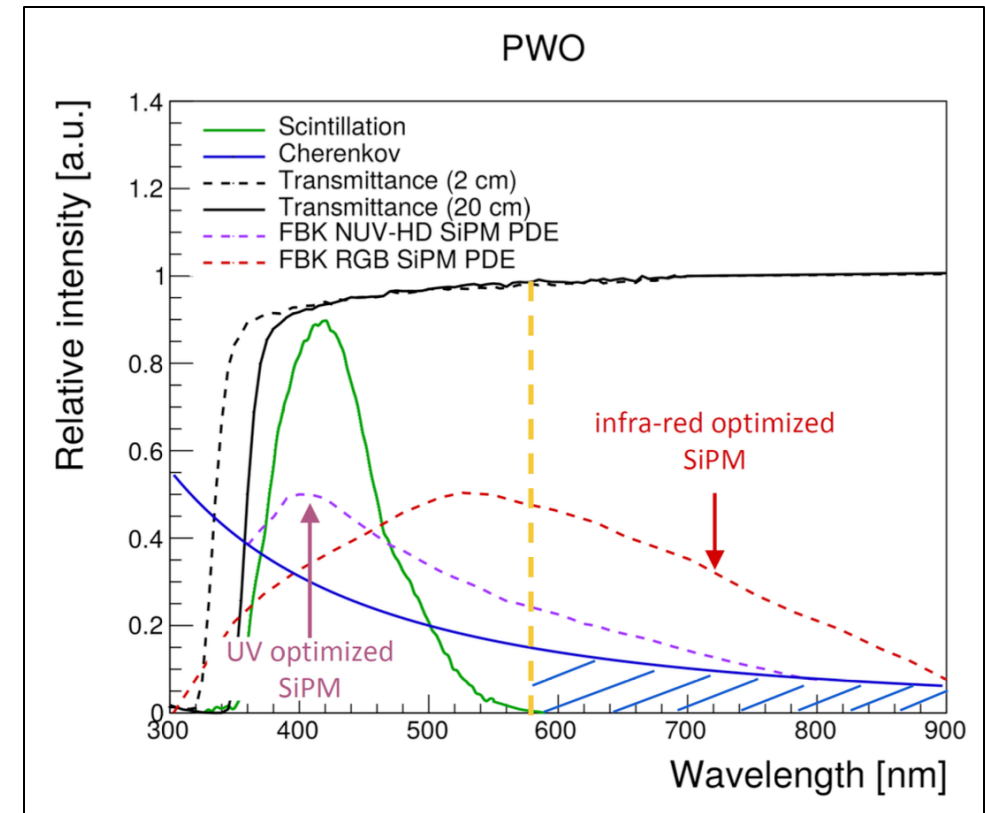
WHY HOMOGENEOUS CRYSTALS FOR DR?

Requirements:

- High density (compact showers)
- High refractive index (Cherenkov yield)
- Good transparency window (UV–visible)
- Sufficient scintillation yield

Prominent candidates:

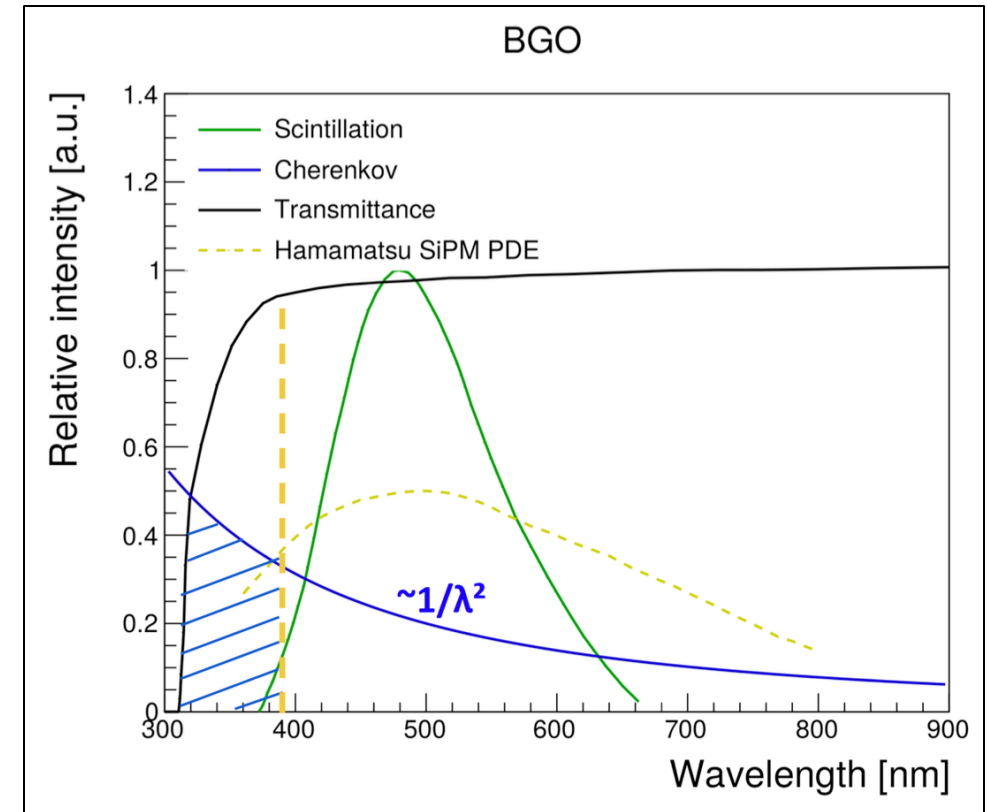
- Lead Tungstate(PbWO_4)
- Bismuth Germanate(BGO)
- Bismuth Silicon Oxide(BSO)



Source:arXiv:2008.00338

WHY BGO AS A DUAL-READOUT CRYSTAL?

- High index of refraction $n \sim 2.15$: strong Cherenkov yield
- Broad transparency region: down to 330nm
- Scintillation:
 - Peak at $\sim 480\text{nm}$: More UV window for Cherenkov signal
 - Long Decay Time ($\sim 300\text{ns}$): Excellent time separation
 - High light yield (8–10k photons/MeV)
- Industrially available (SICCAS), affordable (compared to BSO)



Source:arXiv:2008.00338

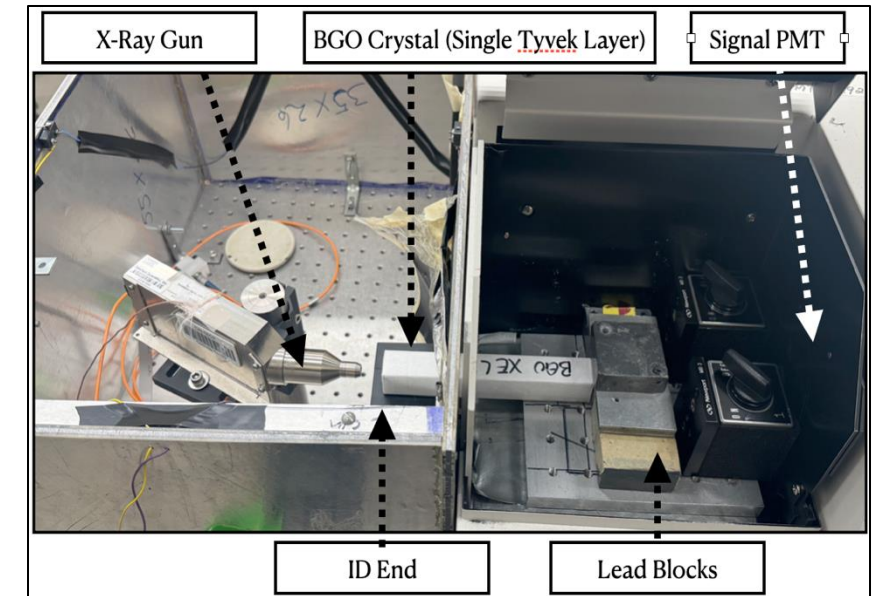
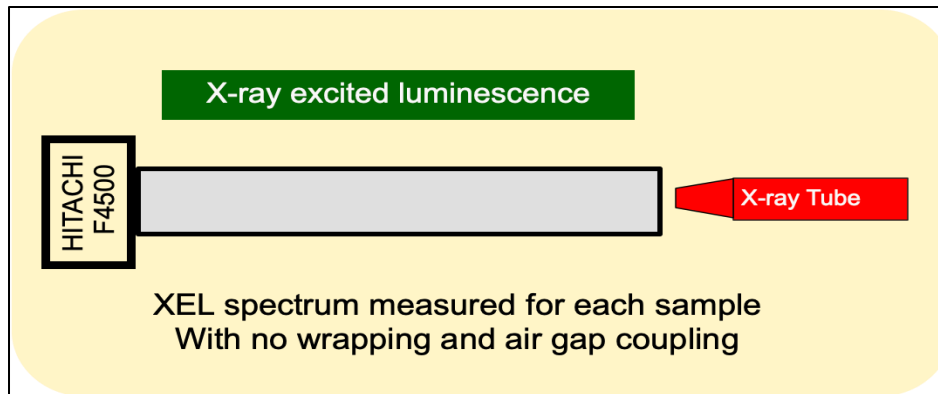
GOALS OF THIS WORK

To validate BGO Samples as DR candidate:

1. X-ray Excited Luminescence (XEL)
 - Measure scintillation peak and emission width
2. Transmission Measurements
 - Direct bandpass transparency
 - Angular dependence (systematics)
 - Cross-facility comparison (CUA vs Caltech)
3. Light Yield Studies
 - Scintillation decay time
 - Light response uniformity
 - Coincidence triggering
4. Commissioning Next-Generation Light Yield Setup

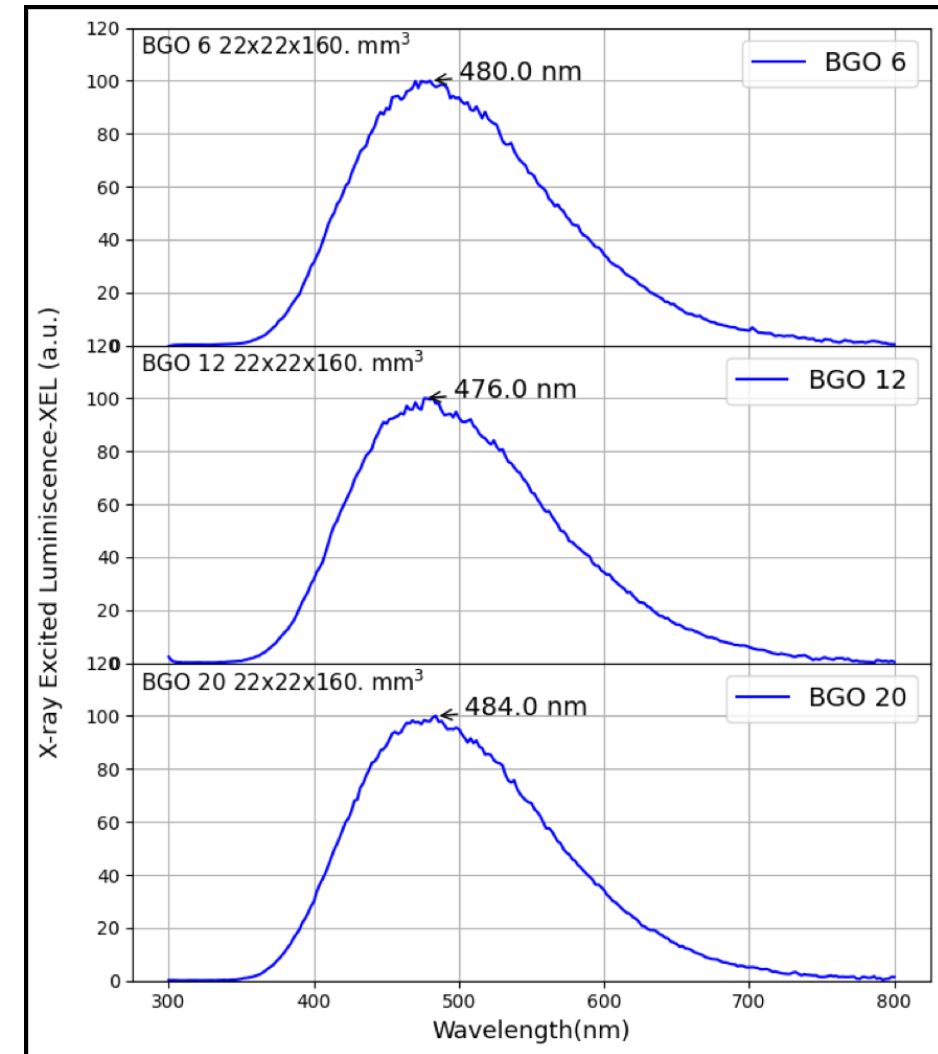
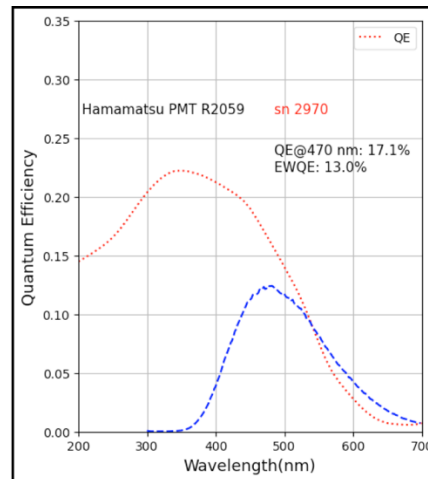
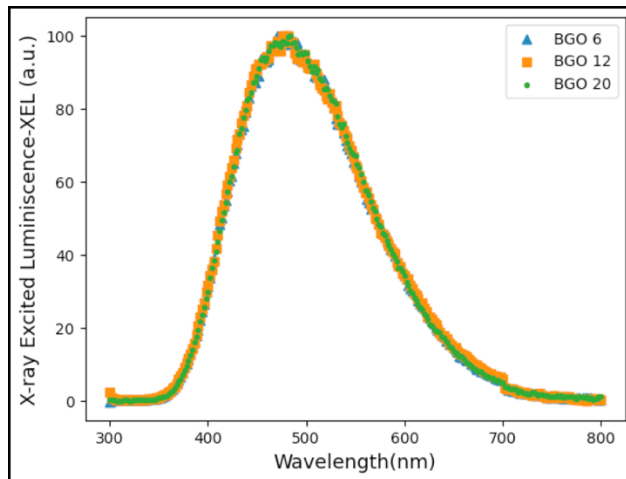
XEL: SETUP AT CALTECH

- X-ray tube + spectrometer
- Acquisition for three BGO samples
- Identify emission peak relevant for DR separation



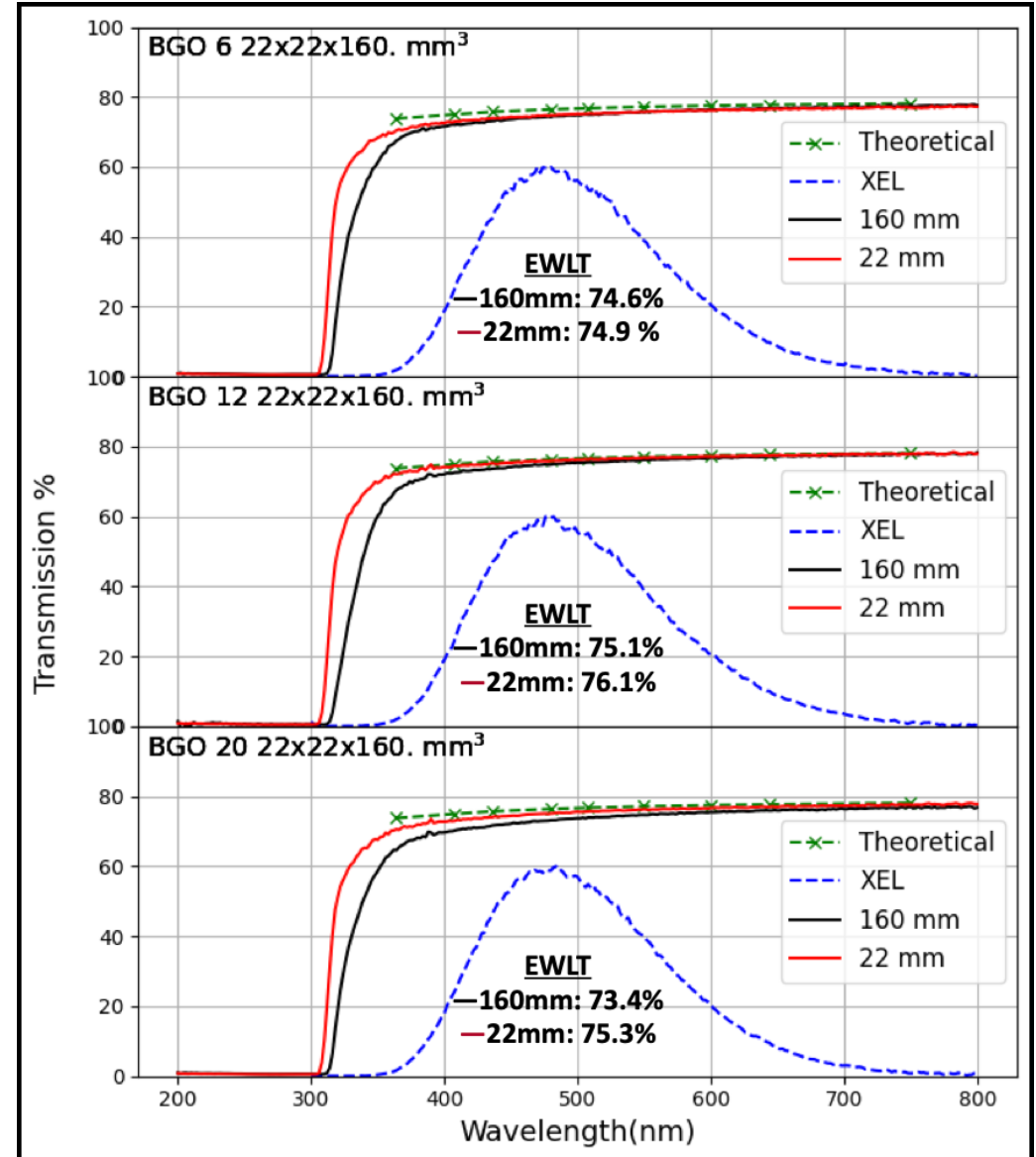
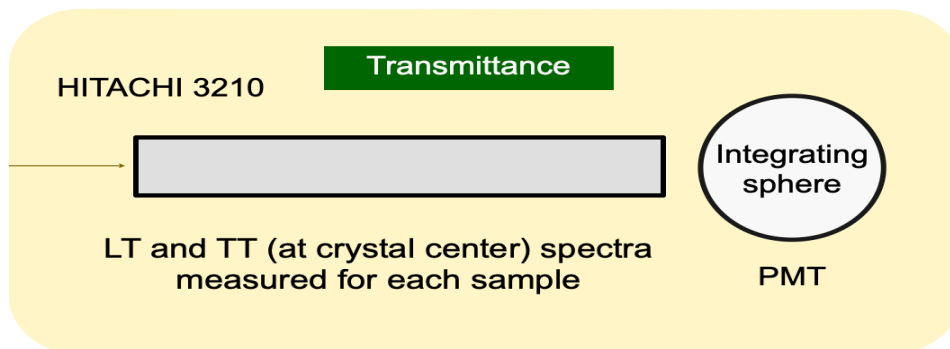
XEL: RESULTS

- Peak around ~ 480 nm -- Good agreement with literature
- Wide emission tail up to 700 nm (helps Cherenkov filtering)
- Sample-to-sample stability is good



TRANSMISSION MEASUREMENTS (CALTECH)

- Excellent transparency in the scintillation region.
 - EWLT~75%
- UV transparency down to 330–350 nm
- Meets DR requirements

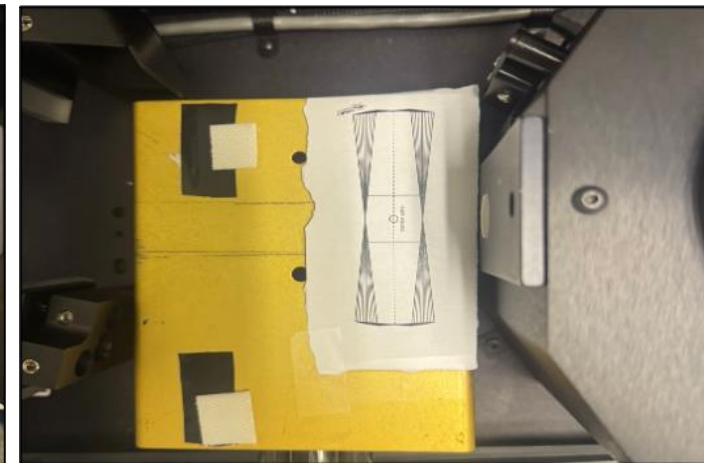
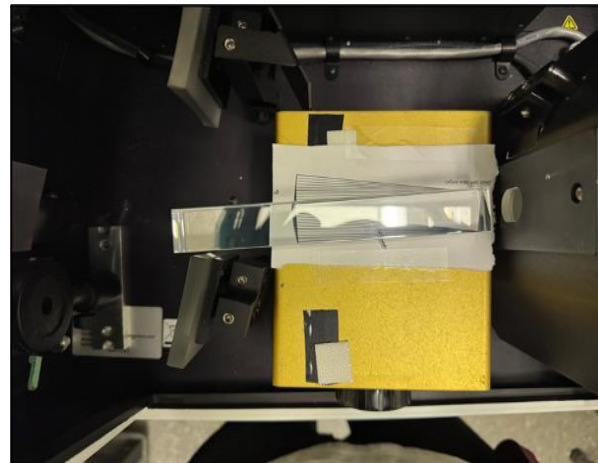
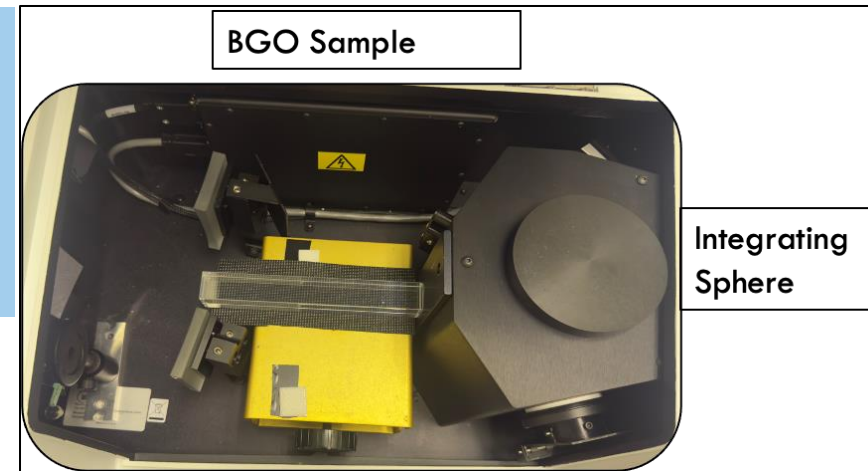


TRANSMISSION MEASUREMENT AT CUA-NP

Measuring:

- Longitudinal vs transverse
- Effect of reseating (repeatability)
- Effect of angular alignment (-5° to $+5^\circ$)
- Comparison with theoretical Fresnel + absorption model
- Cross-comparison Caltech $\leftrightarrow$ CUA

Sample Chamber of
Perkin Elmer Lambda
950
Spectrophotometer at
CUA-NP



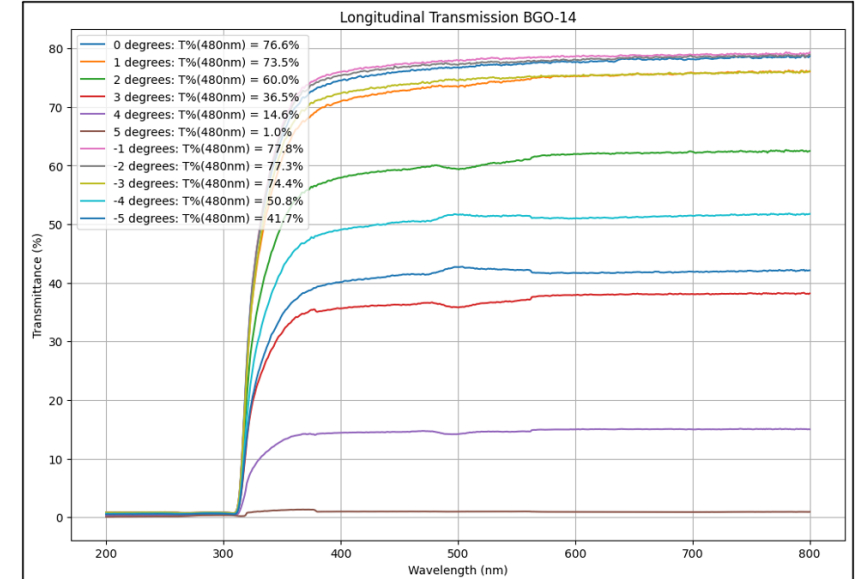
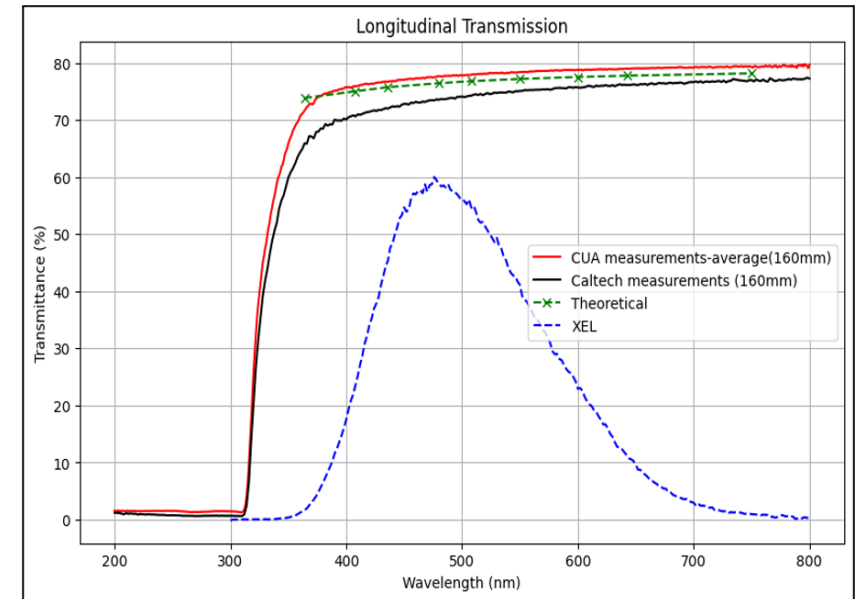
LONGITUDINAL TRANSMISSION RESULTS

Key results:

- Transmission $\sim 78\%$ above 420 nm
- Standard deviation from reseating $\sim 0.7\%$
- Comparable to dual-readout precision needs
- Stronger sensitivity to angle

Observation:

- CUA transmission systematically higher than Caltech $\rightarrow$ later shown to match $\sim 1^\circ$ angular offset effect.



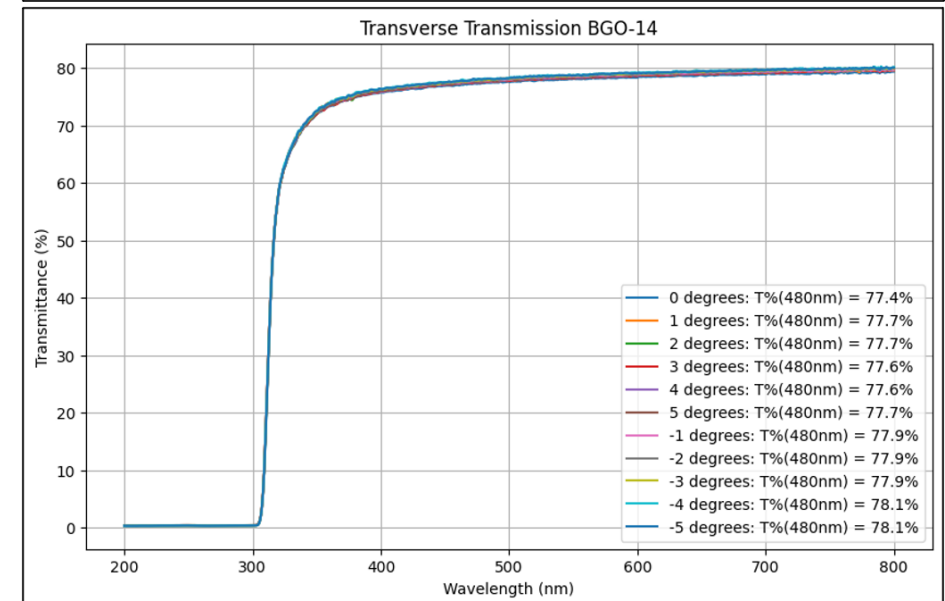
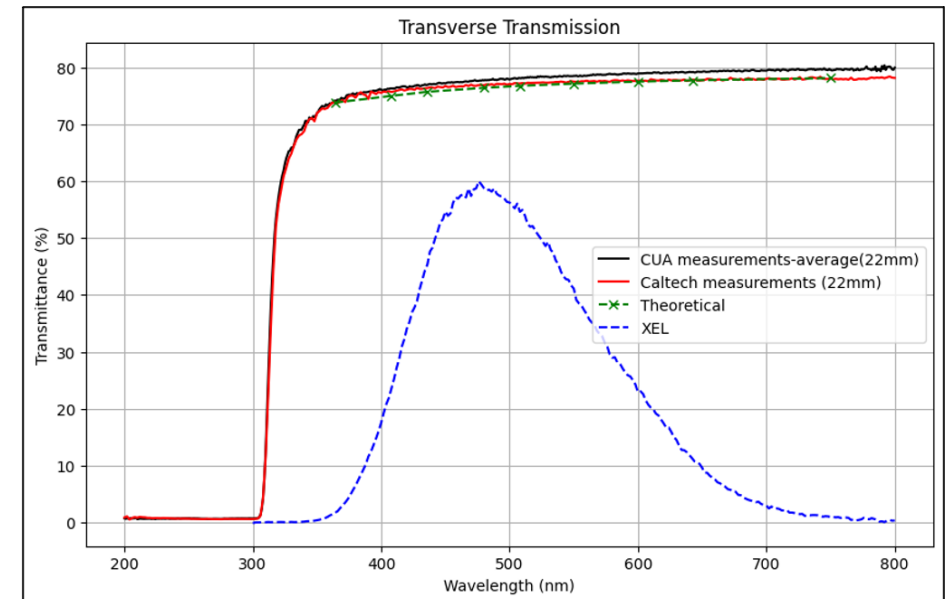
TRANSVERSE TRANSMISSION

Findings:

- Std. dev $\sim 0.4\%$
- Much weaker sensitivity to alignment
- Agreement between CUA & Caltech much closer

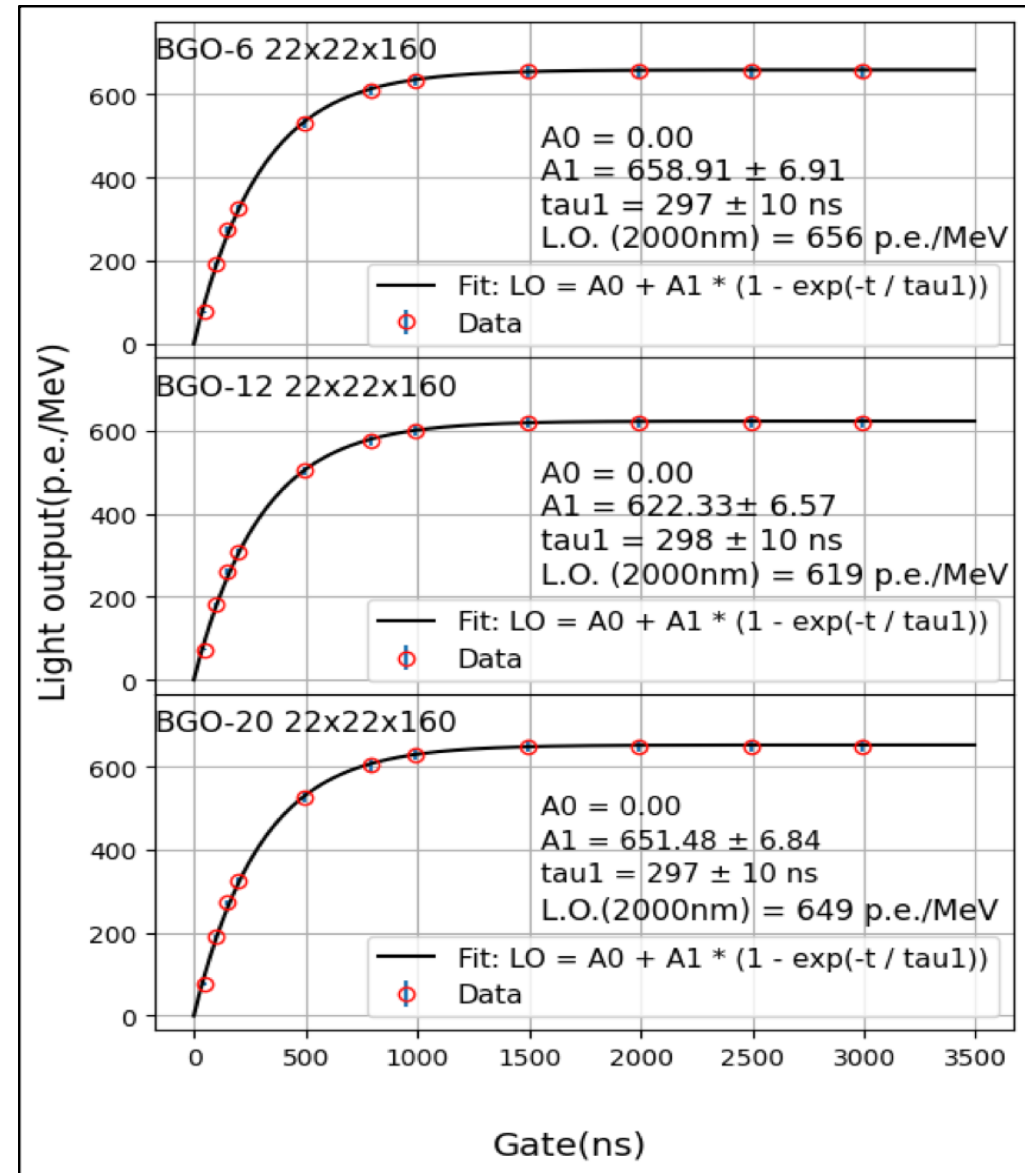
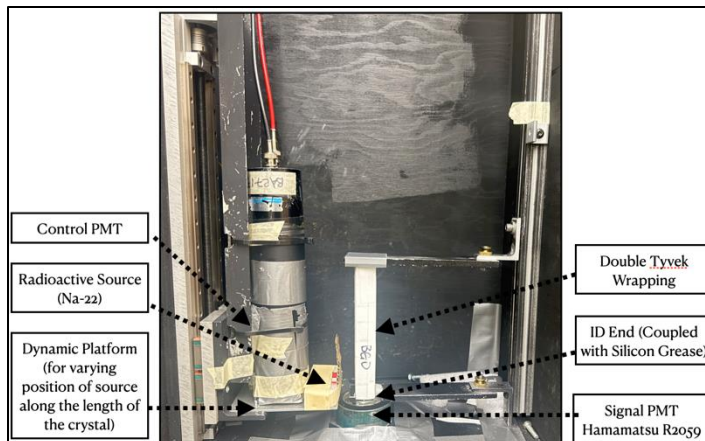
Implication:

- Alignment systematics enter mainly through **long-path geometry**, relevant for calorimetry crystals.



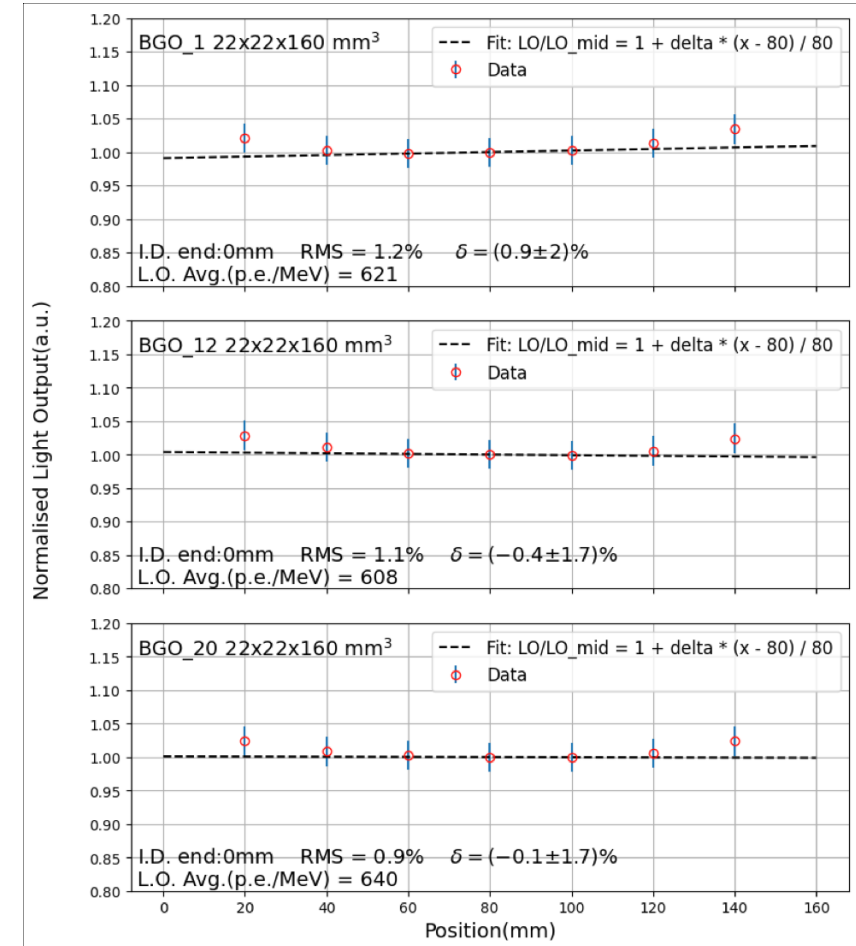
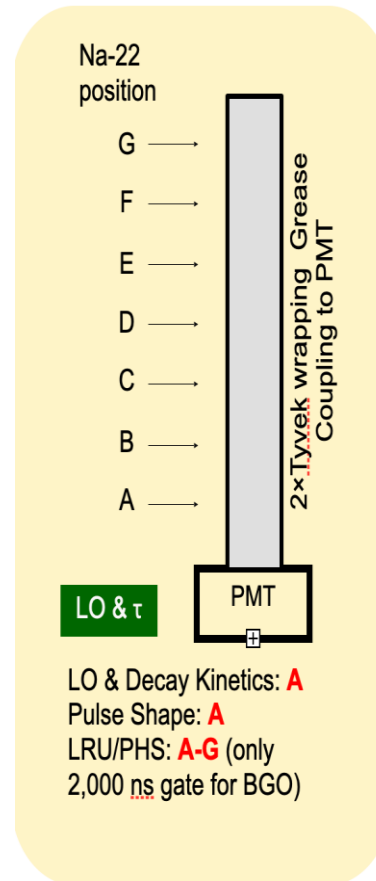
LIGHT YIELD AND DECAY TIME

- Fit produces single-component exponential.
- BGO shows one dominant decay constant (300 ns)
- Important for timing separation vs Cherenkov(<10 ns)



LIGHT RESPONSE UNIFORMITY

- Important for calibration stability
- Stable and uniform response along the crystal length.
- RMS variation of about 1%.



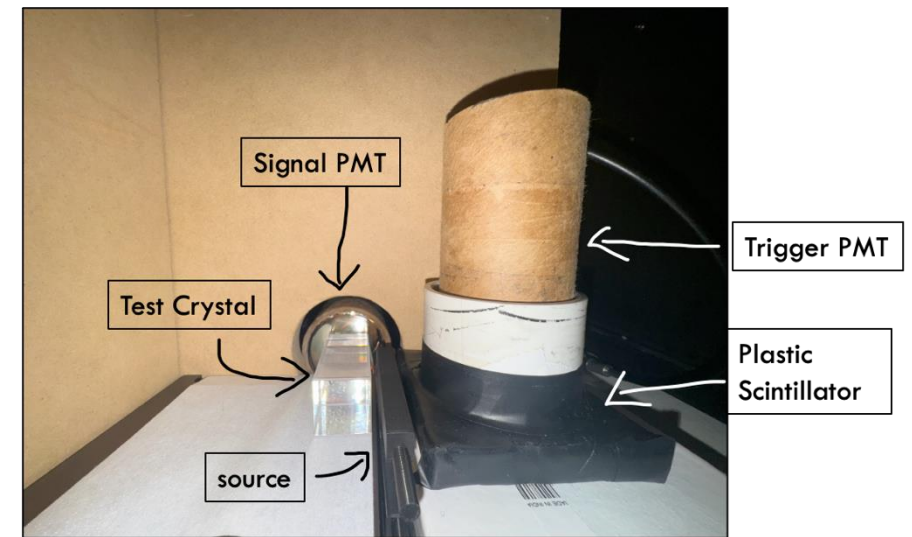
LIGHT-YIELD COMMISSIONING (CUA)

Setup:

- Na-22 source
- Two PMTs (ET 9256B, 900–1000 V)
- Coincidence triggering using plastic scintillator or PWO
- Large BGO ($22 \times 22 \times 160$ mm)

Optical coupling test:

HV	No Grease	With Grease
900 V	5 cpm	39 cpm
1000 V	43 cpm	63 cpm



LIGHT-YIELD COMMISSIONING (UMD)

Setup:

- Cs-137
- Hamamatsu PMT @ 1700 V
- Dark box
- Small BGO (65×22×22 mm)
- Self-triggered sig

Current status:

- Dark-current + cosmic background measured
- Optimizing source–crystal–PMT geometry
- Preparing workflow for the small SICCAS crystals

SUMMARY

- XEL confirms stable scintillation peak at 480 nm
- Transmission excellent across scintillation Cherenkov region
- Systematics in transmission measurements show strong reseating acceptance but also strong angular sensitivity → explains inter-facility shifts
- Light yield measured with:
 - Decay time extraction
 - Uniformity scan
 - Optical coupling studies (large improvement)
- Commissioning of CUA and UMD light yield setup underway

FUTURE PLANS

- Implementing a precision rotation and alignment fixture for transmission measurements to isolate angular biases.
- Repeating transmission scans under controlled warm-up and stabilization protocols to reduce early-run fluctuations.
- Completing coincidence-trigger electronics and data acquisition for the CUA light-yield system.
- Optimizing the source-crystal-PMT geometry at UMD and CUA.
- Extending the analysis to upcoming SICCAS samples with smaller transverse dimensions.

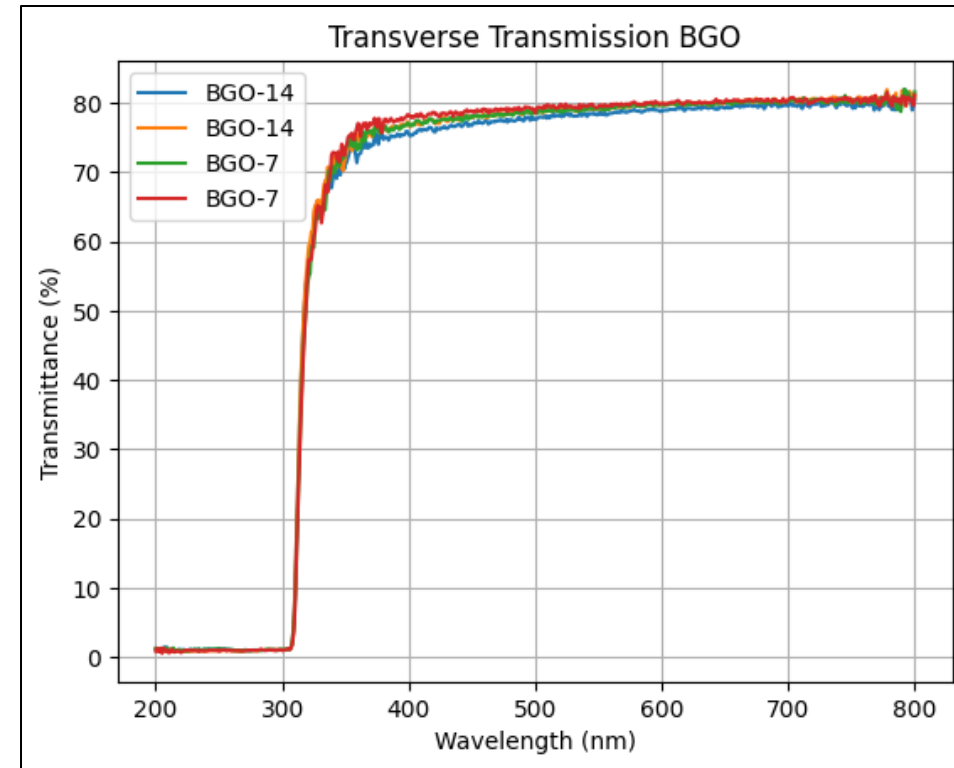
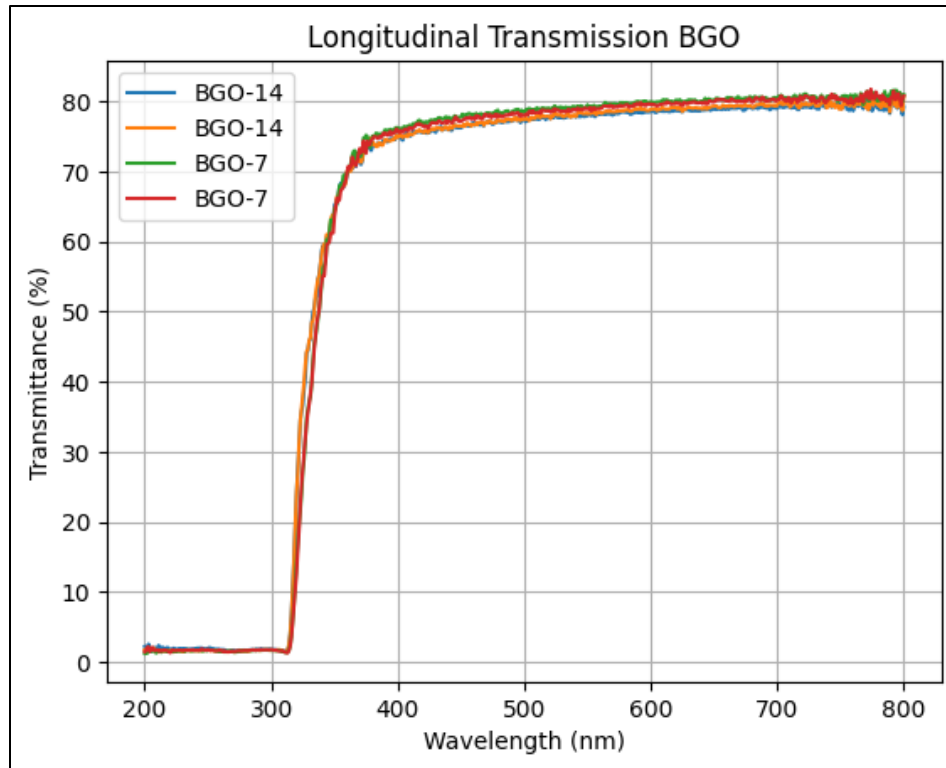
ACKNOWLEDGMENTS

1. CalVision Consortium
2. Prof. H. Newman(Caltech), Prof. R. Yuan (Caltech), Dr. L. Zhang (Caltech)
3. Prof. Sarah Eno(UMD) for assisting with the setup at UMD
4. CUA-NP group for assisting with the Spectrophotometer operations.
5. Prof. Tanja Horn for her Inputs and Guidance.

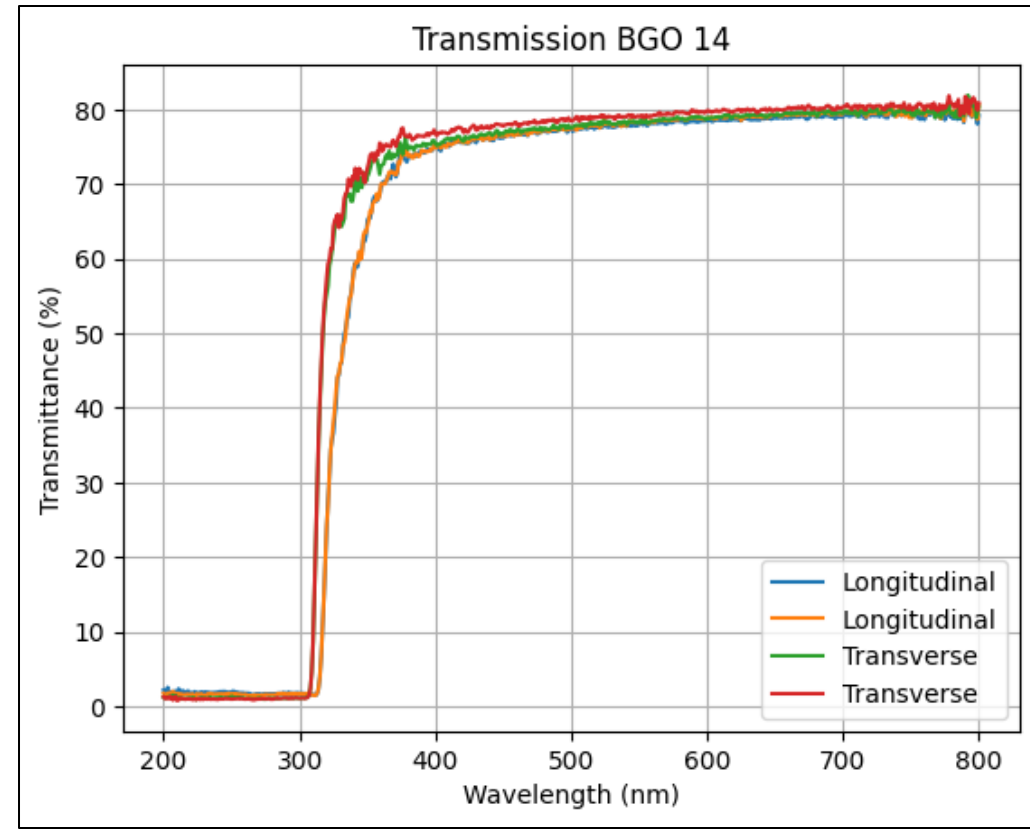
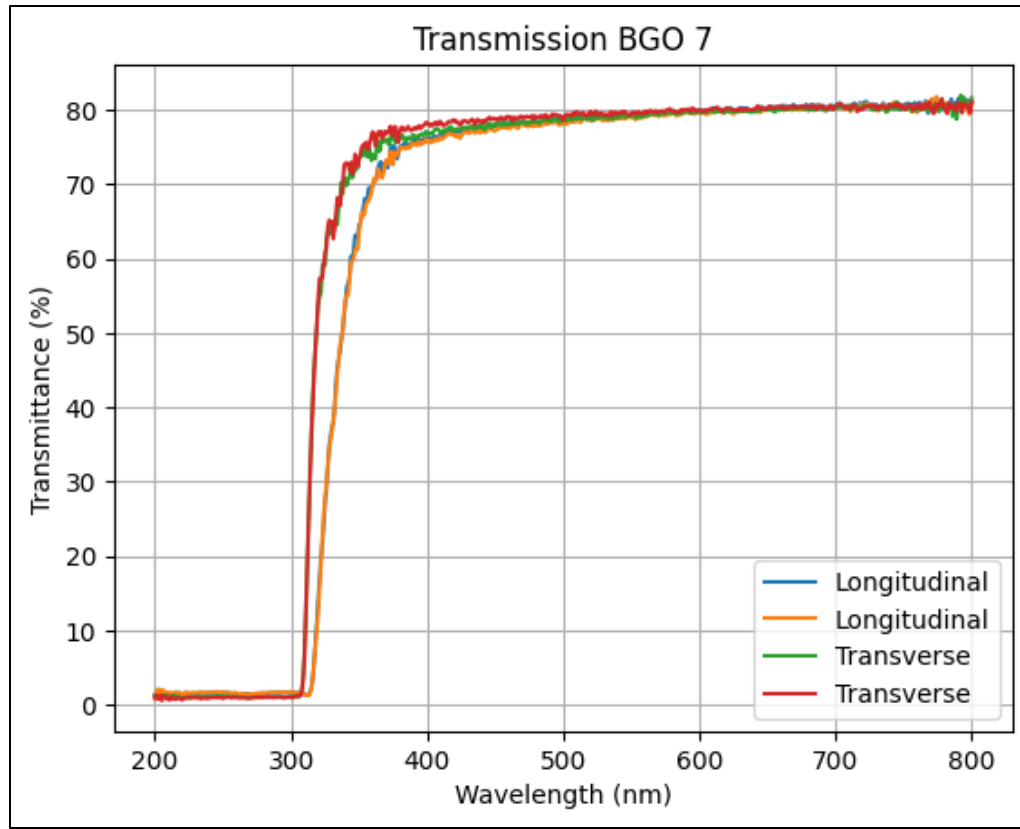
THANK YOU ALL, AND HAPPY HOLIDAYS!



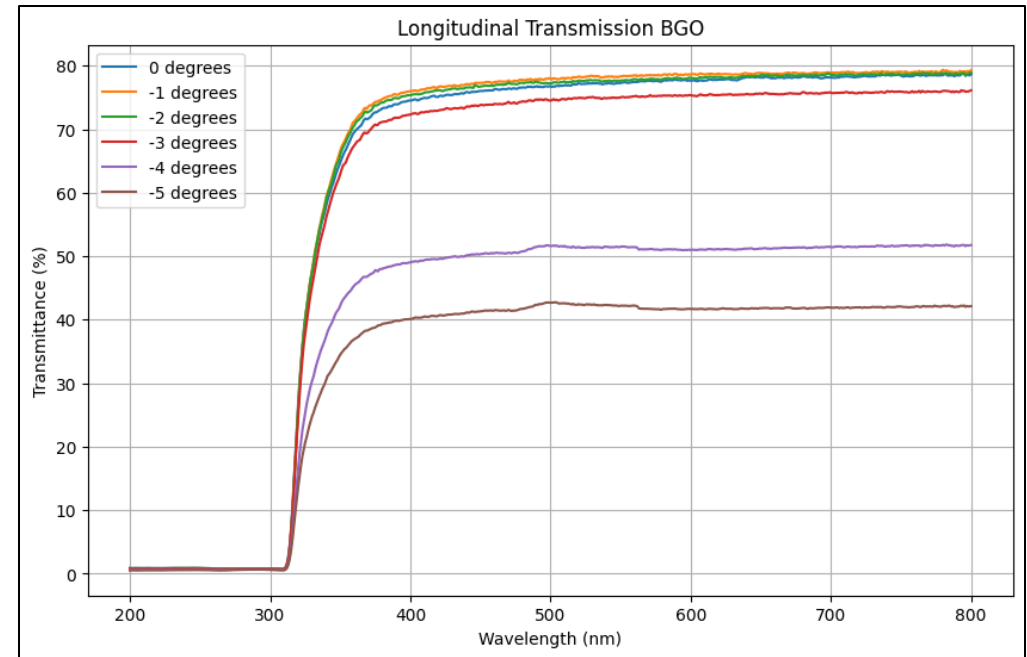
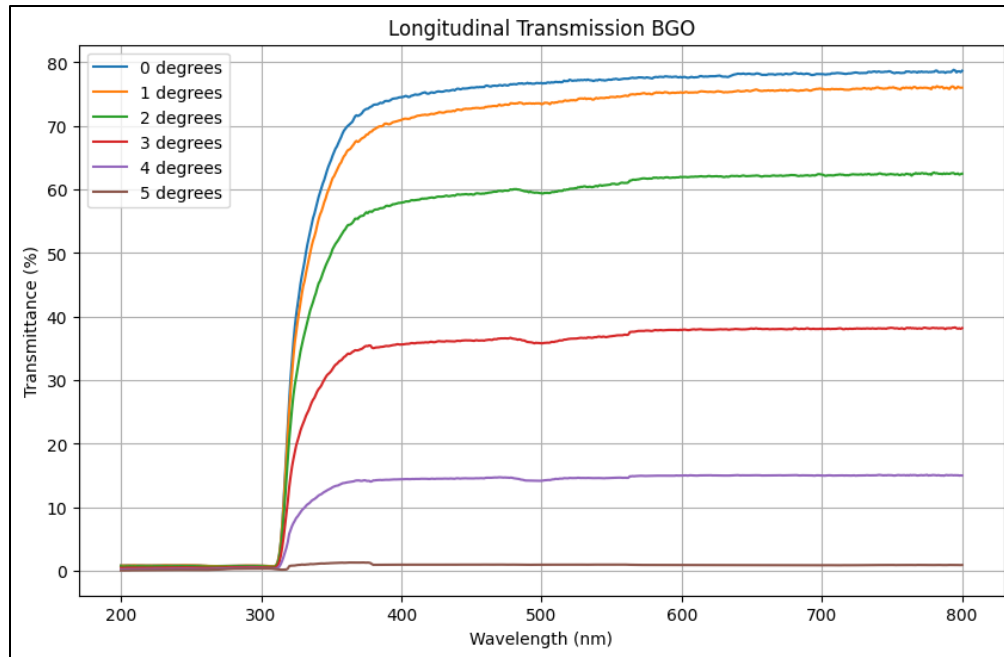
BACKUP



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